Supplemental Material: Orthogonal mode analysis of two-dimensional small-angle neutron scattering

SANS2D mode analyzer contributors

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Abstract

This document describes the analysis implemented in `sans2d-mode-analyzer` (version 0.1.0). A real-valued detector pattern on a circular annulus is represented by orthonormal radial Legendre modes and real angular Fourier modes. We specify the linear fit, its covariance, the associated Parseval powers, angular block statistics and optional numerical Holm diagnostics, and the radial coefficient spectra. We also describe detector selection, optional pixel integration and parity-based centering, and symmetry diagnostics. All quantities refer to the chosen annulus and finite fitted expansion. We finally distinguish numerical verification, data-level validation, and physical interpretation as separate stages of the analysis.

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S1. METHODOLOGY OF THE TWO-DIMENSIONAL SANS MODE ANALYSIS

A. Scope, detector coordinates, and integration measure

The input is a reduced, real-valued scattering intensity or differential cross section, denoted by $I(q, \theta)$. The analysis describes its radial and angular dependence without prescribing a particle form factor or a magnetization model. In particular, the modal fit does not estimate vortex amplitudes, particle radii, or structure-factor parameters. Such physical interpretations require a separate model comparison.

The detector coordinates are q_z along the chosen reference axis and q_y transverse to it. For the magnetic-SANS geometry of the accompanying nanoflower manuscript, the reference axis is the applied-field direction. The code uses

$$q = \sqrt{q_z^2 + q_y^2}, \qquad \theta = \operatorname{atan2}(q_y, q_z), \qquad (q_z, q_y) = q(\cos\theta, \sin\theta). \tag{S1}$$

Angles are understood modulo 2π. The analysis domain and its area are

$$\mathcal{A} = \{(q, \theta) : q_{\min} \le q \le q_{\max},\ 0 \le \theta < 2\pi\}, \qquad A = \pi\Delta, \qquad \Delta = q_{\max}^2 - q_{\min}^2 > 0. \tag{S2}$$

Here $q_{\min} \ge 0$; when the origin is included, its angular coordinate has no effect on continuous integrals because it has zero area.

For a radial-measure exponent $s > -1$, the inner product is

$$\langle f, g \rangle_{\mathcal{A},s} = \int_{q_{\min}}^{q_{\max}} \int_0^{2\pi} f(q, \theta) g(q, \theta)\, \mathrm{d}\theta\, q^s\, \mathrm{d}q. \tag{S3}$$

The default $s = 1$ is the two-dimensional detector-plane area measure $q\, \mathrm{d}q\, \mathrm{d}\theta$. Other exponents intentionally reweight the radial interval; for example, $s > 1$ emphasizes larger q. The choice is not a change of the experimental cross section and should be reported with the analysis.

B. Orthonormal expansion and mode indices

The radial coordinate and radial modes are

$$\Delta_s = q_{\max}^{s+1} - q_{\min}^{s+1},$$
$$x_s(q) = \frac{2q^{s+1} - q_{\max}^{s+1} - q_{\min}^{s+1}}{\Delta_s}, \qquad q^s\,\mathrm{d}q = \frac{\Delta_s}{2(s+1)}\,\mathrm{d}x_s, \tag{S4}$$
$$R_m^{(s)}(q) = \sqrt{\frac{(2m+1)(s+1)}{\Delta_s}}\,P_m[x_s(q)], \qquad m = 0, 1, \ldots, \tag{S5}$$

where P_m is the Legendre polynomial of degree m. Since $\int_{-1}^{1} P_m(x)P_l(x)\,\mathrm{d}x = 2\delta_{ml}/(2m+1)$,

$$\int_{q_{\min}}^{q_{\max}} R_m^{(s)}(q)R_l^{(s)}(q)\,q^s\,\mathrm{d}q = \delta_{ml}. \tag{S6}$$

At the default $s = 1$, these expressions reduce exactly to the coordinate affine in q^2 and the normalization used previously.

To distinguish the angular basis from the block statistic T_n below, we write the angular functions as $\mathcal{T}_n$:

$$\mathcal{T}_n(\theta) = \begin{cases} (2\pi)^{-1/2}, & n = 0, \\ \pi^{-1/2}\cos(n\theta), & n > 0, \\ \pi^{-1/2}\sin(|n|\theta), & n < 0. \end{cases} \qquad \int_0^{2\pi} \mathcal{T}_n\mathcal{T}_{n'}\,\mathrm{d}\theta = \delta_{nn'}. \tag{S7}$$

Thus the signed index n encodes the trigonometric branch; it is not a negative angular frequency of a complex Fourier expansion. The product modes and fitted intensity are

$$\Phi_{nm}^{(s)}(q,\theta) = R_m^{(s)}(q)\mathcal{T}_n(\theta), \qquad \widehat{I}(q,\theta) = \sum_{(n,m)\in\mathcal{K}} \widehat{c}_{nm}\Phi_{nm}^{(s)}(q,\theta), \tag{S8}$$

with $\langle\Phi_{nm}^{(s)}, \Phi_{n'm'}^{(s)}\rangle_{\mathcal{A},s} = \delta_{nn'}\delta_{mm'}$. By default, the standard analyzer uses the full rectangle $|n| \le n_{\max}$ and $0 \le m \le m_{\max}$, containing $K = (2n_{\max}+1)(m_{\max}+1)$ product modes. With `even_only=True`, the fit uses the subset with even $|n|$ at every radial order, so that $\widehat{I}(q,\theta+\pi) = \widehat{I}(q,\theta)$. The lower-level fitting functions accept any explicitly selected set $\mathcal{K}$. Omitted coefficients are understood as zero in subsequent sums.

For a continuous, unweighted orthogonal projection of a square-integrable function, the coefficients would be

$$c_{nm}^{\mathrm{proj}} = \langle I, \Phi_{nm}^{(s)}\rangle_{\mathcal{A},s}. \tag{S9}$$

The implemented fit instead estimates them jointly from the available detector samples. Masks, nonuniform sampling, experimental uncertainties, and finite pixels generally destroy orthogonality of the sampled columns. Consequently, Eq. (S9) is not used as a shortcut for fitting masked or uncertainty-weighted detector data.

C. Sample selection, linear fitting, and covariance

Let $p = 1, \ldots, N$ enumerate the admitted detector samples and j enumerate the selected pairs (n, m). In the default point-evaluation approximation, $X_{pj} = \Phi_j(q_p, \theta_p)$ and $y_p = I_p$. Selection requires finite coordinates and intensity, and a sample center in the chosen annulus. If uncertainties σ_p or sample weights w_p are supplied, their selected values must additionally be finite and strictly positive. Finite zero and negative background-corrected intensities remain in the fit. Logarithmic color scales affect display only.

The objective is

$$Q(\boldsymbol{c}) = (\boldsymbol{y} - X\boldsymbol{c})^{\mathsf{T}} W (\boldsymbol{y} - X\boldsymbol{c}), \qquad W_{pp} = \frac{w_p}{\sigma_p^2}. \tag{S10}$$

Absent sample weights are set to unity. If uncertainties are absent, $\sigma_p = 1$ defines the numerical metric and the residuals estimate its scale. With neither input, minimizing Q is equivalent to minimizing the ordinary sample mean squared error. The solver applies linear least squares to $W^{1/2}X$ and $W^{1/2}\boldsymbol{y}$ using `numpy.linalg.lstsq`; it does not solve a nonlinear optimization problem. A rank-deficient selected design matrix is rejected. For full column rank, the formal solution is

$$\widehat{\boldsymbol{c}} = (X^{\mathsf{T}} W X)^{-1} X^{\mathsf{T}} W \boldsymbol{y}, \qquad \nu = N - K. \tag{S11}$$

For equal-area Cartesian detector pixels, selecting exponent s in the standard analyzer also multiplies the fit weights by

$$\rho_s(q) = \frac{(q_{\max}^2 - q_{\min}^2)/2}{[q_{\max}^{s+1} - q_{\min}^{s+1}]/(s+1)} q^{s-1}. \tag{S12}$$

The constant normalizes the area-average of ρ_s to one and has no effect on the fitted coefficients. Thus the basis orthonormality and detector residual metric use the same $q^s \, \mathrm{d}q \, \mathrm{d}\theta$ measure. Supplied uncertainty weights, sample weights, and fractional coverage multiply this factor. Arbitrary point lists do not by themselves define a quadrature measure; lower-level callers must supply the appropriate sample weights. The reported `mse` always equals

$N^{-1}\sum_p (y_p - \widehat{y}_p)^2$, regardless of the fitting metric. The weighted sum of squares is $Q(\widehat{\boldsymbol{c}})$; the field `reduced_chi_square` equals Q/ν. The latter has a conventional reduced-χ^2 interpretation only under the corresponding noise model.

Let $L = (X^\mathsf{T} W X)^{-1} X^\mathsf{T} W$ denote the linear map from observations to coefficients. With absolute uncertainties, the supplied σ_p are the actual standard deviations of independent pixel measurements, so $\Sigma_{\text{abs}} = \operatorname{diag}(\sigma_p^2)$. Additional sample weights change the fitting metric, not these measurement variances. For $\nu > 0$, the implemented coefficient covariance is

$$C = \begin{cases} L\Sigma_{\text{abs}} L^\mathsf{T}, & \text{absolute pixel uncertainties,} \\ \gamma (X^\mathsf{T} W X)^{-1}, & \text{relative precision model,} \end{cases} \qquad \gamma = Q(\widehat{\boldsymbol{c}})/\nu. \tag{S13}$$

The first case is the default when experimental uncertainties are supplied and implements the diagonal-noise specialization of the general fixed-design sampling covariance

$$C_{\text{sampling}} = (X^\mathsf{T} W X)^{-1} X^\mathsf{T} W \Sigma W X (X^\mathsf{T} W X)^{-1}. \tag{S14}$$

For $w_p = 1$ and $\Sigma = \Sigma_{\text{abs}}$, it reduces to the usual inverse normal matrix. For fractional area weights it generally does not: the middle factor contains w_p^2/σ_p^2, rather than w_p/σ_p^2. Multiplying all geometric weights by a common positive constant leaves both the coefficient estimates and this absolute covariance unchanged. Statistical precision factors must be incorporated in the supplied σ_p themselves.

The covariance is evaluated using a reduced QR factorization $W^{1/2} X = UR$, without forming an inverse normal matrix or an $N \times N$ observation matrix. In the absolute case,

$$L\Sigma_{\text{abs}}^{1/2} = R^{-1} U^\mathsf{T} \operatorname{diag}(\sqrt{w_p}), \qquad C = (L\Sigma_{\text{abs}}^{1/2})(L\Sigma_{\text{abs}}^{1/2})^\mathsf{T}.$$

The relative case uses triangular solves for $\gamma R^{-1} R^{-\mathsf{T}}$.

Relative uncertainties can explicitly request residual scaling; fits without supplied uncertainties also use this second case. It retains the working model $\Sigma = \gamma W^{-1}$, in which the combined fit weights represent relative statistical precision. It does not calibrate geometric coverage weights with otherwise unspecified pixel-noise variance. Merely rescaling residuals does not supply the missing noise model. The implementation returns an undefined covariance when $\nu \leq 0$.

Returned errors and significance thresholds are conditional on the selected noise and linear-signal model. General non-diagonal sample covariances are not accepted; the errors

do not include uncertainty in the fitted center, mode selection, or reduction-induced correlations. For deterministic simulation data, residual-scaled errors describe finite-expansion discrepancies under that model, rather than experimental counting noise.

D. Parseval identities and angular powers

For any coefficient vector in the selected orthonormal basis, including a detector-fitted vector, continuous orthonormality gives the exact identity

$$\mathcal{P} = \|\widehat{I}\|^2_{\mathcal{A},s} = \int_{q_{\min}}^{q_{\max}} \int_0^{2\pi} \widehat{I}^2 q^s \,\mathrm{d}\theta \,\mathrm{d}q = \sum_{(n,m)\in\mathcal{K}} \widehat{c}^{\,2}_{nm}. \tag{S15}$$

For the infinite complete expansion this is Parseval's identity for I; for a finite continuous projection the omitted-mode norm is the sum of the omitted squared projection coefficients. Equation (S15) is exact for the finite reconstructed function even when its coefficients were estimated from irregular or masked data. It does not equate its continuous norm to the measured weighted residual statistic or to the norm of an unknown signal in unobserved regions.

The separate angular powers are

$$\begin{aligned}
\mathcal{P}^c_n &= \sum_{m:(n,m)\in\mathcal{K}} \widehat{c}^{\,2}_{n,m}, && n = 0, 1, \ldots, \\
\mathcal{P}^s_n &= \sum_{m:(-n,m)\in\mathcal{K}} \widehat{c}^{\,2}_{-n,m}, && n = 1, 2, \ldots, \qquad \mathcal{P}^s_0 = 0, \\
\mathcal{P} &= \sum_{n\geq 0} (\mathcal{P}^c_n + \mathcal{P}^s_n).
\end{aligned} \tag{S16}$$

These powers quantify squared-intensity content on the annulus. They are neither integrated scattering cross sections nor magnetic energies, and they discard coefficient signs. For example, with $M_s = 2\pi\Delta_s/(s+1)$ and $\Phi^{(s)}_{00} = 1/\sqrt{M_s}$, the linear integral in the selected measure is $\iint \widehat{I} q^s \,\mathrm{d}\theta \,\mathrm{d}q = \sqrt{M_s}\,\widehat{c}_{00}$ when the constant mode is present. If q has units Q and intensity has units U, then R_m has units $Q^{-(s+1)/2}$, c_{nm} has units $UQ^{(s+1)/2}$, and $\mathcal{P}$ has units U^2Q^{s+1}. Comparisons of absolute modal powers therefore require consistent coordinate units, intensity normalization, and annulus limits.

For a branch block $\widehat{\boldsymbol{c}}^b_n$ and its marginal covariance block C^b_n, with $b \in \{c, s\}$, the plotted power uncertainty is computed as

$$\delta\mathcal{P}^b_n = \left[2\,\mathrm{tr}\big((C^b_n)^2\big) + 4(\widehat{\boldsymbol{c}}^b_n)^{\mathsf{T}} C^b_n \widehat{\boldsymbol{c}}^b_n\right]^{1/2}. \tag{S17}$$

This is the square root of the Gaussian quadratic-form variance evaluated with the fitted coefficient vector, including the term quadratic in covariance. This substitution gives a plug-in uncertainty estimate rather than an exact sampling standard deviation with a known true coefficient vector. Powers are not noise-bias corrected; squaring noisy coefficients produces positive power even for a zero true block. In the standard overview both angular power panels use

$$M = \max_{n,b} \mathcal{P}_n^b, \qquad p_n^b = \frac{\mathcal{P}_n^b}{M}, \qquad \delta p_n^b = \frac{\delta \mathcal{P}_n^b}{M}.$$

The same M is used for both branches and is held fixed in error propagation. The displayed values are not fractions of $\mathcal{P}$. The power-axis label $P_n^2/P_{\max}^2$ in the current implementation denotes $\mathcal{P}_n^b/M$; it does not square that power ratio again. On the logarithmic axes, lower error bars are shortened where necessary to remain above the positive plotting floor; the returned uncertainties retain Eq. (S17). The separate (n, m) power maps instead display $\widehat{c}_{nm}^2/\max_j \widehat{c}_j^2$, using the largest individual product-mode power as their common scale.

E. Radial coefficient spectra associated with angular modes

Keeping each angular order separate while summing its radial modes gives the spectra returned by `angular_radial_spectra`:

$$I_n^c(q) = \sum_m \widehat{c}_{n,m} R_m^{(s)}(q), \quad n \geq 0, \qquad I_n^s(q) = \sum_m \widehat{c}_{-n,m} R_m^{(s)}(q), \quad n \geq 1. \tag{S18}$$

The stored sine spectrum at order zero is identically zero. These are signed coefficient functions with intensity units, and the reconstruction is

$$\widehat{I}(q,\theta) = \frac{I_0^c(q)}{\sqrt{2\pi}} + \frac{1}{\sqrt{\pi}} \sum_{n\geq 1} \left[I_n^c(q)\cos(n\theta) + I_n^s(q)\sin(n\theta) \right]. \tag{S19}$$

Writing the unnormalized real Fourier series as $\widehat{I} = a_0 + \sum_{n\geq 1}[a_n \cos(n\theta) + b_n \sin(n\theta)]$, the Fourier amplitudes and azimuthal average are

$$a_0(q) = \langle \widehat{I}(q,\theta) \rangle_\theta = \frac{I_0^c(q)}{\sqrt{2\pi}}, \qquad a_n(q) = \frac{I_n^c(q)}{\sqrt{\pi}}, \qquad b_n(q) = \frac{I_n^s(q)}{\sqrt{\pi}} \quad (n \geq 1). \tag{S20}$$

Thus I_0^c itself is not the azimuthally averaged intensity. The spectra retain radial sign changes and relative harmonic amplitudes that cannot be recovered from angular powers alone. Radial orthonormality connects the two descriptions exactly:

$$\int_{q_{\min}}^{q_{\max}} [I_n^b(q)]^2 q^s \, \mathrm{d}q = \mathcal{P}_n^b. \tag{S21}$$

In the unnormalized Fourier convention just specified, the same identities read

$$\mathcal{P}_0^c = 2\pi \int_{q_{\min}}^{q_{\max}} a_0(q)^2 q^s \, \mathrm{d}q,$$
$$\mathcal{P}_n^c = \pi \int_{q_{\min}}^{q_{\max}} a_n(q)^2 q^s \, \mathrm{d}q, \qquad \mathcal{P}_n^s = \pi \int_{q_{\min}}^{q_{\max}} b_n(q)^2 q^s \, \mathrm{d}q \quad (n \geq 1).$$

These factors are already included in the orthonormal coefficient convention of Eq. (S16).

Let $\boldsymbol{R}_n^b(q)$ contain the radial modes present in the selected branch block. Covariance propagation gives

$$\mathrm{Var}[I_n^b(q)] = [\boldsymbol{R}_n^b(q)]^{\mathsf{T}} C_n^b \boldsymbol{R}_n^b(q), \qquad \mathrm{Cov}[I_n^b(q), I_n^b(q')] = [\boldsymbol{R}_n^b(q)]^{\mathsf{T}} C_n^b \boldsymbol{R}_n^b(q'). \tag{S22}$$

The standard analyzer evaluates the spectra at 300 equally spaced q values and displays pointwise one-standard-deviation bands from the first expression. This grid samples the fitted function; it introduces no new independent observations. Simultaneous confidence bands are not calculated.

F. Piecewise-linear radial reference analysis

As a reference that does not use global radial polynomials, the same angular modes can be combined with local linear hat functions. Let

$$q_{\min} = q_0 < q_1 < \cdots < q_K = q_{\max} \tag{S23}$$

define K nonoverlapping radial intervals. The nodal functions $B_k(q)$ obey $B_k(q_l) = \delta_{kl}$, are linear on each adjacent interval, and vanish elsewhere. The reference expansion is

$$\widehat{I}_{\mathrm{PL}}(q,\theta) = \sum_n \sum_{k=0}^{K} a_{n,k} B_k(q) \mathcal{T}_n(\theta), \qquad I_{n,\mathrm{PL}}(q) = \sum_{k=0}^{K} a_{n,k} B_k(q). \tag{S24}$$

Hence $a_{n,k} = I_{n,\mathrm{PL}}(q_k)$ is directly the fitted polygonal spectrum at a knot. The knots may be uniform in q or equally spaced in the selected radial measure,

$$q_k = \left[q_{\min}^{s+1} + \frac{k}{K} \left(q_{\max}^{s+1} - q_{\min}^{s+1} \right) \right]^{1/(s+1)}. \tag{S25}$$

The fit uses the same admitted pixels, angular orders, uncertainties, quadrature weights, and optional pixel integration as the polynomial branch. No smoothing penalty is applied.

The hat functions overlap and are not orthogonal. Their coefficients must therefore not be assigned powers by summing $a_{n,k}^2$. Instead define

$$(G_s)_{kl} = \int_{q_{\min}}^{q_{\max}} B_k(q) B_l(q) q^s \, \mathrm{d}q. \tag{S26}$$

The matrix G_s is symmetric and tridiagonal. The reference angular power is

$$\mathcal{P}_{n,\mathrm{PL}}^b = (\boldsymbol{a}_n^b)^\mathsf{T} G_s \boldsymbol{a}_n^b, \tag{S27}$$

which uses the same continuous norm as Eq. (S16). For a coefficient covariance C_n^b, its plug-in Gaussian uncertainty is

$$\delta\mathcal{P}_{n,\mathrm{PL}}^b = \left[2\,\mathrm{tr}(G_s C_n^b G_s C_n^b) + 4(\boldsymbol{a}_n^b)^\mathsf{T} G_s C_n^b G_s \boldsymbol{a}_n^b\right]^{1/2}. \tag{S28}$$

Agreement between the polynomial and polygonal $I_n(q)$ and angular powers tests whether an inferred radial feature is robust to the chosen radial representation.

G. Angular block statistics T_n

Angular powers measure amplitude, whereas `angular_block_tests` tests whether an entire angular contribution is distinguishable from zero under the fit covariance model. For each branch and order the null is $H_{0,n}^b : \boldsymbol{c}_n^b = \mathbf{0}$, with every other fitted coefficient left free. The statistic and block rank are

$$T_n^b = (\widehat{\boldsymbol{c}}_n^b)^\mathsf{T} (C_n^b)_\tau^+ \widehat{\boldsymbol{c}}_n^b, \qquad r_n^b = \#\{j : s_j > \tau_n^b\}, \tag{S29}$$

where r_n^b is the numerical rank. For a covariance block of dimension k_n^b, a single singular-value decomposition determines both this rank and the numerical pseudoinverse:

$$C_n^b = U\,\mathrm{diag}(s_j) V^\mathsf{T}, \qquad \tau_n^b = k_n^b \epsilon_{\mathrm{mach}} s_{\max},$$
$$(C_n^b)_\tau^+ = V\,\mathrm{diag}(d_j) U^\mathsf{T}, \qquad d_j = \begin{cases} s_j^{-1}, & s_j > \tau_n^b, \\ 0, & s_j \le \tau_n^b. \end{cases}$$

Here ϵ_{mach} is the machine precision of the double-precision covariance array. The same retained-singular-value mask is used for the rank and pseudoinverse; values exactly at the cutoff are discarded. The latter is the Moore–Penrose pseudoinverse of the truncated covariance block and reduces to the ordinary inverse when all directions are retained. All correlations between the fitted radial coefficients enter the covariance block; individual squared

coefficient-to-error ratios cannot in general replace Eq. (S29). The quantities T_n^b are dimensionless test statistics, not expansion coefficients or Parseval powers.

With independent Gaussian errors of known absolute scale and the propagated sampling covariance in Eq. (S13), the null reference distribution is $\chi^2_{r_n^b}$, including for additional geometric fit weights. When the common variance scale is estimated from the same fit residuals, the reference is $T_n^b/r_n^b \sim F_{r_n^b,\nu}$ under the relative precision model $\Sigma = \gamma W^{-1}$. This F reference is not calibrated for arbitrary geometric weights with a different noise model. Thus the critical value returned at confidence β is

$$T^b_{n,\mathrm{crit}} = \begin{cases} \chi^2_{r_n^b;\,\beta}, & \text{absolute covariance scale,} \\ r_n^b F_{r_n^b,\nu;\,\beta}, & \text{residual-estimated covariance scale.} \end{cases} \tag{S30}$$

The default is $\beta = 0.95$; upper-tail probabilities are also returned. In the residual-scaled case the code evaluates the upper tail of the $F_{r_n^b,\nu}$ distribution at T_n^b/r_n^b; the stored statistic itself is not divided by r_n^b. The standard analyzer selects this case when `covariance_scaled_by_residual` is true and passes the fit's residual degrees of freedom to the block-test function. Missing blocks have rank zero and undefined statistics, including the nonexistent zero-order sine block. A nominal rejection concerns the specified block conditional on the other included modes, without correcting for the other blocks tested. A small amplitude can have a large T_n^b if it is precisely determined. The noise-model qualifications following Eq. (S13) apply to these tests.

For nearly singular blocks the statistic tests only the covariance directions retained by the common cutoff. The same numerical rank determines the reference distributions and thresholds above. An existing block with zero numerical rank has a zero stored statistic and undefined upper-tail probability and critical value; it provides no calibrated test.

The block-test result additionally provides an optional numerical Holm step-down correction for one joint family $\mathcal{H}$ containing all positive-rank cosine and sine blocks for this map, fixed annulus, and selected basis. The family size is $N_{\mathrm{test}} = |\mathcal{H}|$; missing and rank-zero blocks are excluded. For the full rectangle with $n_{\max} = 25$ and positive block ranks, $N_{\mathrm{test}} = 51$, independent of the radial truncation. Both branches are corrected together, including the isotropic block, rather than defining a separate family for each panel.

Let $p_{(1)} \le \cdots \le p_{(N_{\mathrm{test}})}$ denote the ordered uncorrected upper-tail probabilities. The

adjusted probabilities, mapped back to their original blocks after sorting, are

$$\widetilde{p}_{(j)} = \min\left\{1,\ \max_{1\le i\le j}\left[(N_{\mathrm{test}} - i + 1)p_{(i)}\right]\right\}, \qquad j = 1, \ldots, N_{\mathrm{test}}. \tag{S31}$$

The rejection rule is $\widetilde{p}_n^b \le \alpha$, where $\alpha = 1 - \beta = 0.05$ by default. Unavailable probabilities in positive-rank blocks count conservatively as one in the adjustment and remain undefined without rejection in the output. Missing and rank-zero blocks are excluded.

Provided the marginal probabilities are valid, Holm controls the probability of rejecting any true null in this family by α under arbitrary dependence. It does not repair an incorrect covariance model, account for data-dependent basis or annulus selection, or control a larger family spanning several maps or analysis runs. The T_n^b statistics, numerical ranks, raw probabilities, and nominal critical values remain unchanged.

These error probabilities concern zero-coefficient null hypotheses under the specified noise and signal model. They do not quantify the frequency of experimental artifacts or physical misidentifications. A centering or detector artifact can produce a genuinely nonzero fitted block and hence a small model-based p-value. Calibration simulations check the implementation for their generating models; they do not establish that those assumptions hold for the measured data.

The standard overview shows the T_n^b values but deliberately omits critical-value curves and rejection markers. The nominal critical values, raw probabilities, and numerical Holm properties remain available separately; they do not determine a physical identification criterion, modify the modal powers and spectra, or select and refit a reduced model.

H. Parity, mirror symmetry, and signal centering

Point reflection in the detector plane is $\theta \mapsto \theta + \pi$. Both trigonometric branches obey $\mathcal{T}_n(\theta + \pi) = (-1)^{|n|}\mathcal{T}_n(\theta)$. An exactly parity-even pattern therefore has only even angular orders. The package reports

$$f_{\mathrm{odd}} = \frac{\sum_{|n|\,\mathrm{odd},m} \widehat{c}_{nm}^{\,2}}{\mathcal{P}}, \qquad f_{\mathrm{even}} = 1 - f_{\mathrm{odd}}, \tag{S32}$$

provided $\mathcal{P} > 0$. For the fitted expansion, Parseval gives

$$D_{\mathrm{par}}^2 = \frac{1}{M_s} \int_{q_{\min}}^{q_{\max}} \int_0^{2\pi} [\widehat{I}(q,\theta) - \widehat{I}(q,\theta+\pi)]^2 q^s \,\mathrm{d}\theta \,\mathrm{d}q = \frac{4\mathcal{P} f_{\mathrm{odd}}}{M_s}, \tag{S33}$$

$$d_{\mathrm{par}} = \frac{\|\widehat{I}(q,\theta) - \widehat{I}(q,\theta+\pi)\|_{\mathcal{A},s}}{\|\widehat{I}\|_{\mathcal{A},s}} = 2\sqrt{f_{\mathrm{odd}}}. \tag{S34}$$

The first measure has intensity-squared units; the second is a scale-free relative RMS mismatch and contains the same information as f_{odd}. These are diagnostics of the fitted function. With `even_only=True`, the odd power and parity mismatch vanish by construction. Mirror symmetry about the q_z axis is $\theta \mapsto -\theta$ and forbids all sine coefficients; it is an additional assumption independent of parity.

For an observable expected to be parity even, a broad background of odd angular powers can serve as an empirical reference for measurement noise, asymmetries, and processing artifacts. With the same annulus and power normalization, an even power at or below that level is not clearly resolved above this empirical background. This is a descriptive robustness criterion, not a calibrated probability threshold: mode variances, radial profiles, and sensitivities can differ. Physical odd contributions must first be excluded before interpreting odd powers as a background.

For a cross section known to be parity even, an incorrect origin can create odd harmonics. The optional centering step estimates a common displacement of the reciprocal-space origin from antipodal intensity differences. For a two-dimensional affine detector grid, the fitted coordinate map is

$$\begin{pmatrix} q_z \\ q_y \end{pmatrix} = B \begin{pmatrix} r \\ s \end{pmatrix} + \boldsymbol{b}, \qquad \boldsymbol{\delta} = B \begin{pmatrix} \delta_r \\ \delta_s \end{pmatrix}, \qquad \boldsymbol{q}_p' = \boldsymbol{q}_p - \boldsymbol{\delta}, \tag{S35}$$

where r, s are row and column indices. The antipodal partner in the original coordinates is $2\boldsymbol{\delta} - \boldsymbol{q}_p$. Bilinear interpolation gives $\widetilde{I}(2\boldsymbol{\delta} - \boldsymbol{q}_p) = \sum_j a_j I_j$ and conditional variance $\widetilde{\sigma}_p^2 = \sum_j a_j^2 \sigma_j^2$. All neighbors with nonzero interpolation weight must be valid. The minimized objective is

$$Q_{\mathrm{cen}}(\boldsymbol{\delta}) = \sum_{p \in \mathcal{H}(\boldsymbol{\delta})} \frac{[I_p - \widetilde{I}(2\boldsymbol{\delta} - \boldsymbol{q}_p)]^2}{\sigma_p^2 + \widetilde{\sigma}_p^2}. \tag{S36}$$

Here $\mathcal{H}(\boldsymbol{\delta})$ selects valid partners within the corrected annulus and one representative half-plane, avoiding explicit antipodal double counting. The admissible set can change with the trial shift. The implementation minimizes the unnormalized sum, records the before/after

objectives and pair counts, and uses a bounded Powell search. The standard analyzer bounds each detector-index displacement by two pixels by default; the standalone grid refiner defaults to one pixel. Multiple images can be passed to the standalone refiner to minimize the sum of their objectives for one common shift.

An optional second, modular refinement operates directly on the modal representation. For every trial reciprocal-space shift $\boldsymbol{\delta}$, the data are refitted on a separately chosen finite basis $\mathcal{K}_{\mathrm{cen}}$, and the minimized objective is

$$J_{\mathrm{odd}}(\boldsymbol{\delta}) = \frac{\sum_{\substack{(n,m)\in\mathcal{K}_{\mathrm{cen}} \\ |n| \text{ odd}}} \widehat{c}_{nm}(\boldsymbol{\delta})^2}{\sum_{(n,m)\in\mathcal{K}_{\mathrm{cen}}} \widehat{c}_{nm}(\boldsymbol{\delta})^2}. \tag{S37}$$

The standalone function `refine_center_by_odd_mode_power` accepts an explicit mode list and a bounded shift in reciprocal-space units. The standard analyzer exposes it through `odd_mode_centering=True`; by default it uses the compact full rectangle $|n| \le 5$, $0 \le m \le 8$ and limits the additional shift to 0.25 nominal pixels around the current origin. It can start from the nominal origin or locally refine the pixelwise parity result. The fitting basis, annulus, measure exponent, sample weights, before/after objective and power, selected-point counts, number of evaluations, and recovered shift are retained as diagnostics. The final scientific fit remains independent of the centering basis.

For a one-dimensional point list, the center is estimated on a regular 129×129 auxiliary grid spanning $[-q_{\max}, q_{\max}]^2$ by default. Valid source points within $1.05q_{\max}$ are used for linear intensity interpolation; centering uncertainties are assigned from the nearest measured point. After refining this grid, the displacement is applied to the original point coordinates. The final modal fit retains the original intensities, uncertainties, and point sampling. In either centering path, the selected annulus is reevaluated after the shift. Centering requires supplied uncertainties and is disabled by default in the standard analyzer. Geometry-dependent externally supplied weights must be recomputed for the shifted origin; internal pixel integration computes coverage after centering.

Interpolation correlations and center-estimation uncertainty are not propagated. Accordingly, neither Q_{cen} nor J_{odd} is a calibrated goodness-of-fit statistic for the original data. The modal optimum is additionally conditional on $\mathcal{K}_{\mathrm{cen}}$, the annulus, weights, and search bounds. A physical odd contribution must not be absorbed into a parity-based center. In particular,

the polarization-averaged spin-flip cross section and an individual spin-flip channel need not have the same parity because an individual channel can contain a chiral contribution. An odd-power fraction alone therefore does not identify a reduction or centering error.

I. Finite pixels, rebinning, and explicit preprocessing

For an affine detector grid, optional subpixel quadrature estimates the annular coverage of a pixel P_p and the admitted-area average of each mode:

$$f_p = \frac{|P_p \cap \mathcal{A}|}{|P_p|} \simeq \frac{N_{p,\mathrm{in}}}{s_{\mathrm{sub}}^2}, \qquad \overline{X}_{pj} \simeq \frac{1}{N_{p,\mathrm{in}}} \sum_{\ell \in P_p \cap \mathcal{A}} \Phi_j(q_\ell, \theta_\ell). \tag{S38}$$

Subpixel nodes lie at the centers of a regular $s_{\mathrm{sub}} \times s_{\mathrm{sub}}$ subdivision. The affine map accounts for pixel shape and orientation. There are two distinct uses: `fractional_annulus_weights` supplies f_p as a fit weight, while `pixel_integrated_annulus_design` also replaces the design row by $\overline{X}_{pj}$. The latter admits any pixel with $f_p > 0$ and multiplies an existing sample weight by f_p for $s = 1$. For general s, its returned quadrature weight is

$$w_{p,s} \simeq \frac{1}{s_{\mathrm{sub}}^2} \sum_{\ell \in P_p \cap \mathcal{A}} \rho_s(q_\ell), \tag{S39}$$

which combines fractional coverage and the radial factor in Eq. (S12). The default standard analyzer instead uses center-point selection and evaluation; supplying only external fractional weights retains that center-point annulus selection. One must therefore enable internal pixel integration to include pixels whose centers lie outside the annulus but whose areas overlap it. Supplying the same coverage externally as well would apply it twice.

Pixel averaging refers to the portion inside the analysis annulus; the observed pixel intensity itself is not subdivided. This is an approximation for boundary pixels that represent full-pixel averages. Pixel integration also does not model wavelength spread or instrumental q resolution. With absolute errors, the supplied pixel variances remain unchanged by coverage; Eq. (S13) propagates them using the integrated design and the effective fit weights. For a point list without supplied pixel geometry, these helpers reduce to binary annulus selection and point evaluation. The integrated design is assembled in chunks to bound temporary storage.

The optional `cartesian_bin_average` routine converts point samples into equal-area

Cartesian cells by

$$I_P = \frac{\sum_{\ell \in P} v_\ell I_\ell}{\sum_{\ell \in P} v_\ell}. \tag{S40}$$

For a source grid uniform in (q, θ), $v_\ell = q_\ell$ supplies the Jacobian of $q\,\mathrm{d}q\,\mathrm{d}\theta$; general nonuniform grids require their actual quadrature weights. A duplicated angular endpoint at 2π should be removed before averaging, as in the documented NuMagSANS data preparation. Empty cells remain undefined. The routine returns occupancies and source-weight sums, but does not propagate measurement uncertainties. Equal-area Cartesian samples require no additional factor of q in the subsequent area-based fit.

Further preprocessing functions are explicit optional operations. `symmetry_diagnostic` compares original pixels to bilinearly interpolated parity or mirror partners using residuals standardized by $\sqrt{\sigma_p^2 + \widetilde{\sigma}_p^2}$. Its summed statistic uses one representative half-plane and is conditional on interpolation assumptions. `symmetry_fold` averages a complete symmetry orbit $\mathcal{O}_p$ by inverse variance,

$$I_p^{\mathrm{fold}} = \frac{\sum_{j \in \mathcal{O}_p} I_j/\sigma_j^2}{\sum_{j \in \mathcal{O}_p} 1/\sigma_j^2}, \qquad (\sigma_p^{\mathrm{fold}})^2 = \left(\sum_{j \in \mathcal{O}_p} 1/\sigma_j^2 \right)^{-1}. \tag{S41}$$

Incomplete orbits are marked invalid. Parity folding imposes even orders; adding mirror folding also imposes the absence of sine contributions. The returned per-pixel errors omit correlations from interpolation and shared orbit members. Folding is intended for visualization and sensitivity checks; symmetry-constrained fits to the original samples avoid creating these duplicated observations. Folding is never automatic.

`mask_symmetry_diagnostic` reports a weight-centroid displacement normalized by the largest admitted radius, point-reflection residuals of the mask, and the number of partially covered pixels. `rotate_coordinates` changes the angular origin according to $\theta' = \theta - \varphi$ without changing intensities. `rescale_intensity` divides intensities and uncertainties by a common positive number. `standardized_residual_outliers` flags $|I_p - \widehat{I}_p|/\sigma_p > 5$ by default and does not delete samples. These functions do not implement experimental background, efficiency, or polarization-leakage corrections; those belong to the preceding reduction.

J. Implementation correspondence and data conventions

The definitions above correspond to `basis.py` and `indices.py` (basis and selected modes), `fit.py` (linear fit and covariance), `power.py` (powers and parity), `spectra.py` (radial spectra), and `block_test.py` (angular block tests). All source files lie under `src/sans2d_mode_analyzer/`. Explicit detector operations reside in `preconditioning.py`. The optional polygonal reference is implemented in `piecewise_linear_basis.py` and `piecewise_linear_reference.py`. The `standard_analyzer` combines sample selection, fitting over the selected modes, and all numerical diagnostics. Its optional `radial_measure_exponent` applies the same exponent to the radial basis, detector-fit quadrature weights, Parseval diagnostics, parity normalization, and reported radial spectra; the default is $s = 1$. It returns the fit, detector-space arrays, powers, block tests, parity measures, and radial spectra independently of figure generation. The optional overview is rendered separately by `figure_backend.py`; setting `export_figure=False` suppresses this rendering without changing the numerical analysis. Its final two rows show the signed spectra $I_n^{c,s}(q)$ on linear axes and the corresponding magnitudes $|I_n^{c,s}(q)|$ on common log-log axes. The latter representation changes only the visualization: it exposes weak contributions over several decades but discards coefficient signs and turns every zero crossing into a downward cusp. No additional fit or power calculation is performed.

The ILL NeXus loader maps the stored `qx1` direction to q_z, `qy1` to q_y, and converts the stored inverse-angstrom coordinates to inverse nanometers by a default factor of ten. For the bundled nanoflower data, the local data documentation identifies the 10 and 30 mT maps as $\frac{1}{2}(I^{+-} + I^{-+})$, whereas the 100 mT and 1 T maps contain the $+-$ channel alone. Their common baseline analysis uses $0.015 \leq q \leq 0.100\,\mathrm{nm}^{-1}$, the supplied uncertainties, $n_{\max} = m_{\max} = 25$, 7×7 subpixel coverage weights, and no parity centering or folding. Their supplied pixel errors are treated as absolute, with covariance propagated through the coverage-weighted fit. The ASCII loader reads columns (q_z, q_y, I, σ) without converting units; the legacy nanoporous-Fe benchmark has an unresolved coordinate-unit caveat in its data README. Its numerical q interval must therefore not be assigned physical units without establishing that provenance.

A reproducible analysis specifies the input channel and reduction, coordinate units and angular origin, annulus, retained mode set, weights and covariance scaling, any centering or

symmetry operations, the truncation settings, and the test confidence and family. The modal description then provides a common representation for measured and simulated detector patterns, with physical interpretation tied to the observable and assumptions used.

S2. VERIFICATION AND INTERPRETATION STRATEGY

A. White-Noise Test Model

This test isolates a particularly simple question: if the true intensity contains only a small number of occupied modes, how much fitted power should appear in all other modes solely because independent white noise was added to the detector pixels? Answering this question provides a controlled reference for the "noise carpet" in a modal power map. It is a verification of the linear fit and its covariance propagation, not a general noise model for reduced experimental SANS data.

a. Synthetic data and repeated observations. Let $\boldsymbol{c}$ denote the known coefficient vector used to generate a synthetic detector image and let X be the same design matrix as in Eq. (S11). One noisy observation is

$$\boldsymbol{y} = X\boldsymbol{c} + \boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_N), \tag{S42}$$

where every admitted detector pixel has the same noise variance and different pixels are statistically independent. Repeating the simulation means keeping X, $\boldsymbol{c}$, and σ fixed while drawing a new $\boldsymbol{\epsilon}$ for every realization. The modes used to generate the exact image are called *occupied*; all remaining fitted modes have a true coefficient equal to zero and are called *unoccupied*. The latter are the positions at which the noise floor is examined.

For ordinary least squares, the fitted coefficients satisfy

$$\widehat{\boldsymbol{c}} = \boldsymbol{c} + A\boldsymbol{\epsilon}, \qquad A = (X^{\mathsf{T}}X)^{-1}X^{\mathsf{T}}, \tag{S43}$$

$$C = \mathrm{Cov}(\widehat{\boldsymbol{c}}) = \sigma^2 AA^{\mathsf{T}} = \sigma^2 (X^{\mathsf{T}}X)^{-1}. \tag{S44}$$

Thus the covariance is not obtained from the Monte Carlo calculation: it is known analytically from the imposed pixel noise and the sampled basis. The Monte Carlo calculation independently checks whether repeated numerical fits follow this prediction. Equation (S44) is the equal-variance specialization of Eq. (S13).

b. Why a zero coefficient has nonzero fitted power. The index k below denotes one flattened product-mode index (n, m). Its fitted coefficient is normally distributed,

$$\widehat{c}_k \sim \mathcal{N}(c_k, C_{kk}). \tag{S45}$$

The displayed power of that product mode is the square $p_k = \widehat{c}_k^2$. Squaring removes the sign of a noise fluctuation and therefore produces a positive average even when the true coefficient is zero:

$$\mathrm{E}[p_k] = c_k^2 + C_{kk}, \tag{S46}$$

$$\mathrm{Var}(p_k) = 2C_{kk}^2 + 4c_k^2 C_{kk}. \tag{S47}$$

For an unoccupied mode, $c_k = 0$, and these expressions simplify to

$$\mathrm{E}[p_k] = C_{kk}, \qquad \mathrm{Var}(p_k) = 2C_{kk}^2, \qquad \frac{p_k}{C_{kk}} \sim \chi_1^2. \tag{S48}$$

The diagonal covariance entry C_{kk} is therefore the predicted mean noise power at position (n, m). It is also the quantity labelled "analytic mean" in the Stage-2 power map. The "Monte Carlo mean" is instead the arithmetic mean of $\widehat{c}_k^2$ over many independently generated detector images. Agreement between these two maps verifies the expected transfer of white pixel noise into modal power.

Equation (S48) also explains why a single noisy fit does not look like a perfectly flat carpet. For one unoccupied mode the standard deviation of p_k is $\sqrt{2}$ times its mean. Individual squared coefficients consequently fluctuate strongly, even though their ensemble mean is known exactly. After averaging R independent realizations, the relative standard error of that mean is $\sqrt{2/R}$. For $R = 1000$ this is about 4.5% for each individual mode.

c. When is the carpet constant? White pixel noise does not by itself imply identical coefficient variances. The expected floor is constant only when the diagonal entries C_{kk} are approximately equal. Finite Cartesian sampling, an annular mask, missing pixels, nonuniform uncertainties, or an ill-conditioned set of sampled modes can make these entries different. In the current Stage-2 test the predicted zero-mode powers vary by only about 2% around their mean, so describing the ensemble-averaged result as an approximately constant carpet is appropriate. This near constancy is a property of this design matrix, rather than a general theorem about white noise.

d. Powers summed over several radial modes. The angular powers in Eq. (S16) sum several squared coefficients. For a block $\mathcal{B}$ with coefficient subvector $\widehat{\boldsymbol{c}}_{\mathcal{B}}$ and covariance block $C_{\mathcal{B}}$, the corresponding power $p_{\mathcal{B}} = \widehat{\boldsymbol{c}}_{\mathcal{B}}^{\mathsf{T}}\widehat{\boldsymbol{c}}_{\mathcal{B}}$ obeys

$$\mathrm{E}[p_{\mathcal{B}}] = \boldsymbol{c}_{\mathcal{B}}^{\mathsf{T}}\boldsymbol{c}_{\mathcal{B}} + \mathrm{tr}(C_{\mathcal{B}}), \tag{S49}$$

$$\mathrm{Var}(p_{\mathcal{B}}) = 2\,\mathrm{tr}(C_{\mathcal{B}}^2) + 4\boldsymbol{c}_{\mathcal{B}}^{\mathsf{T}}C_{\mathcal{B}}\boldsymbol{c}_{\mathcal{B}}. \tag{S50}$$

For a true zero block, its mean noise power is therefore $\mathrm{tr}(C_{\mathcal{B}})$ and its variance is $2\,\mathrm{tr}(C_{\mathcal{B}}^2)$. Coefficient correlations enter through the latter trace. In the special case of M independent coefficients with a common variance v, one has $p_{\mathcal{B}}/v \sim \chi^2_M$ and a relative standard deviation $\sqrt{2/M}$. Summed angular powers can consequently appear substantially smoother than the powers of individual (n, m) modes.

e. Numerical realization of the test. The reproducible script `verification_examples/stage_2/example_1.py` uses the deterministic Stage-1 signal, containing four occupied even-angular modes, and fits the full rectangle $|n| \leq 8$, $0 \leq m \leq 8$. It adds Gaussian pixel noise with σ equal to 3% of the maximum absolute exact intensity and evaluates 1000 independent noise realizations. In the resulting (n, m) maps, positive n denotes cosine modes, negative n denotes sine modes, and the four white cells mark the occupied generating modes that are deliberately excluded from the zero-mode comparison. The current test gives an empirical-to-analytic mean-power ratio of 1.0010 and an empirical-to-analytic power-variance ratio of 1.0017. The normalized powers of the unoccupied modes also follow the χ^2_1 density in Eq. (S48). These agreements verify the coefficient covariance and the positive noise bias of squared modal amplitudes under the stated assumptions.

No Monte Carlo calculation is required to obtain Eqs. (S46)– (S50); it is used here as a transparent numerical verification. Experimental data can contain counting-statistical heteroscedasticity, background-subtraction errors, interpolation artifacts, pixel-to-pixel correlations, centering uncertainty, and model truncation error. In those cases the white-noise test remains a useful regression test, but its analytic carpet must not be interpreted as a calibrated experimental significance threshold unless the corresponding experimental covariance model has independently been justified.

B. Heteroskedastic Gaussian Test Model

The Stage-3 test removes the equal-variance assumption while retaining independent Gaussian detector errors. Its purpose is to verify that known, pixel-dependent experimental uncertainties are propagated through the weighted fit correctly. The observation model is

$$\boldsymbol{y} = X\boldsymbol{c} + \boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \Sigma), \qquad \Sigma = \mathrm{diag}(\sigma_1^2, \ldots, \sigma_N^2). \tag{S51}$$

The errors remain statistically independent, but their variances need not be equal. This distinction is important: heteroskedastic noise is not white noise, even though both models have a diagonal pixel covariance.

a. Weighted estimator and analytic covariance. When the supplied standard deviations are the true measurement errors and no additional sample weights are used, inverse-variance weighting gives

$$W = \Sigma^{-1},$$

$$\widehat{\boldsymbol{c}} = \boldsymbol{c} + L\boldsymbol{\epsilon}, \qquad L = (X^{\mathsf{T}}\Sigma^{-1}X)^{-1}X^{\mathsf{T}}\Sigma^{-1}, \tag{S52}$$

$$C = \mathrm{Cov}(\widehat{\boldsymbol{c}}) = L\Sigma L^{\mathsf{T}} = (X^{\mathsf{T}}\Sigma^{-1}X)^{-1}. \tag{S53}$$

This is the inverse-variance specialization of the sandwich covariance in Eq. (S14). The final equality in Eq. (S53) relies on using the correct $W = \Sigma^{-1}$. If the supplied uncertainties do not describe the actual pixel variances, the fit remains numerically defined but its reported covariance and probabilistic calibration need not be correct.

Because Eq. (S51) is Gaussian and the estimator is linear,

$$\widehat{c}_k \sim \mathcal{N}(c_k, C_{kk}), \qquad c_k = 0 \; \mathit{Longrightarrow} \; \frac{\widehat{c}_k^2}{C_{kk}} \sim \chi_1^2. \tag{S54}$$

Thus the normalized zero-mode power has exactly the same reference distribution as in Eq. (S48). What changes is the unnormalized mean power C_{kk}: unequal pixel errors alter the fitted-mode covariance through the complete sampled basis and generally produce a mode-dependent noise floor. Dividing every mode by one common empirical floor would therefore discard relevant information. The appropriate normalization is the individual analytic variance C_{kk}.

b. Controlled radial uncertainty profile. To make this effect visible without introducing a physical SANS count model, the Stage-3 example prescribes a simple radial standard-

deviation profile. For an admitted pixel p, define

$$u_p = \frac{q_p - q_{\min}}{q_{\max} - q_{\min}}, \qquad \sigma_p = 0.015 I_{\max}(1 + 3u_p), \qquad I_{\max} = \max_p |I_p^{\text{exact}}|. \tag{S55}$$

The pixel standard deviation therefore rises from 1.5% to 6% of the maximum exact intensity between the inner and outer annulus boundaries. This profile is deliberately chosen as a transparent numerical test; it is not claimed to be a detector or counting-statistical model.

The script `verification_examples/stage_3/example_1.py` uses the same four-mode exact signal and full $|n| \le 8$, $0 \le m \le 8$ fit as Stage 2. It propagates Eq. (S55) analytically and compares the result with 1000 independent noise realizations. The expected powers of the 149 unoccupied modes have a relative variation of about 10.0%, compared with about 2% in the equal-variance Stage-2 design. Nevertheless, the empirical-to-analytic mean-power ratio is 0.9982, the corresponding power-variance ratio is 0.9943, and the normalized powers follow the χ^2_1 reference distribution. The structured raw power floor is therefore predicted by the weighted design rather than indicating occupied physical modes.

c. Regression criterion. The automated test `tests/test_stage_3_regression.py` generates 1024 independent realizations with a fixed random seed and passes the known σ_p values to the absolute-uncertainty fit. It verifies the reported coefficient covariance against Eq. (S53). For each unoccupied mode it then forms

$$z_{rk} = \frac{\widehat{c}_{rk} - c_k}{\sqrt{C_{kk}}}, \tag{S56}$$

where r labels the realization, and checks $\mathrm{E}[z_k] = 0$, $\mathrm{E}[z_k^2] = 1$, $\mathrm{Var}(z_k^2) = 2$, and 95% coverage of the χ^2_1 quantile. The acceptance tolerances are five analytic Monte Carlo standard errors for 1024 realizations. They are therefore tied to the finite ensemble size rather than to the particular values generated by the fixed seed. This regression test verifies the complete path from a supplied heteroskedastic error map through weighted fitting to calibrated modal power; it does not establish that an experimental error map is itself correct.

C. Correlated Gaussian Test Model

Stage 4 retains Gaussianity but allows different detector pixels to fluctuate coherently. This is the first verification stage for which a map of marginal pixel errors is not sufficient to determine the coefficient covariance. A computationally convenient low-rank model is

$$\boldsymbol{\epsilon} = D\boldsymbol{z} + H\boldsymbol{a}, \qquad \boldsymbol{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_N), \qquad \boldsymbol{a} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_R), \qquad \Sigma = DD^{\mathsf{T}} + HH^{\mathsf{T}}, \tag{S57}$$

where D is diagonal and the R columns of H are fixed spatial background patterns with independent random amplitudes. The two Gaussian vectors are mutually independent. This factorized construction permits sampling and covariance propagation without forming a dense $N \times N$ pixel covariance.

The marginal error reported for pixel p is

$$\sigma^2_{p,\mathrm{marg}} = \Sigma_{pp} = D^2_{pp} + \sum_{r=1}^{R} H^2_{pr}. \tag{S58}$$

These numbers describe the diagonal of Σ but contain no information about its off-diagonal entries. Two noise models can therefore have identical error bars at every pixel and nevertheless produce very different fitted-mode uncertainties.

a. True and diagonal coefficient covariance. The current standard analyzer accepts independent pixel uncertainties rather than a general pixel covariance. When it is supplied only with the marginal errors in Eq. (S58), it uses

$$\Sigma_{\mathrm{diag}} = \mathrm{diag}(\Sigma), \qquad W = \Sigma^{-1}_{\mathrm{diag}}, \qquad L_d = (X^{\mathsf{T}} W X)^{-1} X^{\mathsf{T}} W. \tag{S59}$$

The resulting coefficient estimate remains linear and unbiased for a zero-mean error, but two distinct covariance matrices must be distinguished:

$$C_{\mathrm{reported}} = L_d \Sigma_{\mathrm{diag}} L_d^{\mathsf{T}}, \tag{S60}$$

$$C_{\mathrm{true}} = L_d \Sigma L_d^{\mathsf{T}} = L_d (DD^{\mathsf{T}} + HH^{\mathsf{T}}) L_d^{\mathsf{T}}. \tag{S61}$$

Only C_{true} describes repeated measurements generated by Eq. (S57). For a true zero coefficient,

$$\frac{\widehat{c}_k^2}{(C_{\mathrm{true}})_{kk}} \sim \chi^2_1, \qquad \frac{\widehat{c}_k^2}{(C_{\mathrm{reported}})_{kk}} \sim r_k \chi^2_1, \qquad r_k = \frac{(C_{\mathrm{true}})_{kk}}{(C_{\mathrm{reported}})_{kk}}. \tag{S62}$$

Consequently, a diagonal error model can produce either understated or overstated mode significance. In particular, a smooth correlated pattern can project almost entirely onto one low angular order, even though its marginal pixel variance appears modest.

If the full positive-definite Σ were known, the efficient correlation-aware estimator would instead be generalized least squares,

$$\widehat{\boldsymbol{c}}_{\mathrm{GLS}} = (X^{\mathsf{T}} \Sigma^{-1} X)^{-1} X^{\mathsf{T}} \Sigma^{-1} \boldsymbol{y}, \qquad C_{\mathrm{GLS}} = (X^{\mathsf{T}} \Sigma^{-1} X)^{-1}. \tag{S63}$$

Stage 4 does not add a dense-covariance GLS interface to the standard analyzer. It characterizes the failure of the existing diagonal statistical model when the data-generating process is correlated.

b. Controlled correlated background. The script `verification_examples/stage_4/example_1.py` uses the same exact signal and modal rectangle as Stages 2 and 3. Its independent component has $D_{pp} = 0.020 I_{\max}$. The two columns of H are

$$h_{1,p} = 0.010 I_{\max}, \qquad h_{2,p} = 0.030 I_{\max} \frac{q_{y,p}}{q_{\max}}. \tag{S64}$$

The first is a common offset and the second is a smooth vertical gradient. The gradient is parity odd and therefore couples strongly to sine modes of odd angular order. These deliberately simple patterns are diagnostic test cases, not a proposed instrumental model for a particular experiment.

For 1000 realizations, the empirical-to-true mean-power ratio of the 149 unoccupied modes is 0.9973 and the corresponding power-variance ratio is 1.0018. Normalization by C_{true} therefore reproduces the analytic Gaussian result. In contrast, for the most affected mode $(n, m) = (-1, 0)$ the true coefficient variance is 674.2 times the variance reported by the diagonal model. Its nominal 95% coverage is 94.9% with the true covariance but only 5.9% with the diagonal covariance. The resulting apparently significant odd mode is generated entirely by the correlated background.

The automated test `tests/test_stage_4_regression.py` repeats this experiment with a smaller design and 1024 realizations. It verifies the χ^2_1 moments and coverage under C_{true}, confirms that the public fit reports Eq. (S60), and requires the correlation-induced miscalibration to remain detectable. This is a regression characterization of a known model limitation, rather than a claim that marginal experimental error bars determine the required full covariance.

D. Detector-Centre Translation Test

Stage 5 isolates a deterministic geometrical systematic rather than another noise distribution. The synthetic generator accepts `center_shift=(delta_q_z, delta_q_y)` and evaluates the prescribed modal signal about that displaced physical centre while retaining laboratory detector coordinates. This makes it possible to generate a signal whose true expansion contains only even angular orders, analyze it about an incorrect nominal origin, and then repeat the same analysis after the parity-based centre refinement of Eq. (S36). No random noise is required for this test. Unit uncertainties merely define equal numerical

weights and do not represent a statistical error model.

a. First-order angular-mode coupling. Let the parity-even signal contain only the two unnormalized Fourier components

$$I_{\mathrm{even}}(q,\theta) = A_0(q) + A_2(q)\cos(2\theta). \tag{S65}$$

If its physical centre is at $\boldsymbol{\delta} = (\delta_z, \delta_y)$ but the pattern is analyzed about the origin, the observed detector function is

$$I_{\boldsymbol{\delta}}(q_z, q_y) = I_{\mathrm{even}}(q_z - \delta_z, q_y - \delta_y) = I_{\mathrm{even}} - \delta_z \partial_{q_z} I_{\mathrm{even}} - \delta_y \partial_{q_y} I_{\mathrm{even}} + O(|\boldsymbol{\delta}|^2). \tag{S66}$$

Using

$$\partial_{q_z} = \cos\theta\,\partial_q - \frac{\sin\theta}{q}\partial_\theta, \qquad \partial_{q_y} = \sin\theta\,\partial_q + \frac{\cos\theta}{q}\partial_\theta, \tag{S67}$$

the displacement along q_y gives

$$\begin{aligned}\partial_{q_y} I_{\mathrm{even}} = &\left(A_0' - \frac{1}{2}A_2' - \frac{A_2}{q}\right)\sin\theta \\ &+ \left(\frac{1}{2}A_2' - \frac{A_2}{q}\right)\sin(3\theta),\end{aligned} \tag{S68}$$

where primes denote radial derivatives. Thus a small vertical translation of an exact $n = 0, 2$ cosine signal produces the distinctive odd sine orders $n = 1$ and $n = 3$. The corresponding q_z derivative is

$$\begin{aligned}\partial_{q_z} I_{\mathrm{even}} = &\left(A_0' + \frac{1}{2}A_2' + \frac{A_2}{q}\right)\cos\theta \\ &+ \left(\frac{1}{2}A_2' - \frac{A_2}{q}\right)\cos(3\theta),\end{aligned} \tag{S69}$$

so a horizontal translation couples to the cosine branches of the same odd orders. A general displacement produces both patterns. The minus signs in Eq. (S66) determine the signs of the induced spectra. These relations show why coherent radial profiles in $I_1^{c,s}(q)$ and $I_3^{c,s}(q)$ can diagnose a centre displacement more specifically than a single total odd-power fraction.

b. Controlled numerical realization. The script `verification_examples/stage_5/example_1.py` generates the three exact product modes $(n, m) = (0, 0), (0, 2), (2, 1)$ on $0.05 \le q \le 0.30$ and displaces their common centre by 0.35 nominal pixels along q_y, or $\delta_y = 2.625 \times 10^{-3}$ in the chosen reciprocal-coordinate units. The raw full-rectangle fit has an odd-power fraction $f_{\mathrm{odd}} = 6.9043 \times 10^{-4}$. Its two dominant odd sine powers occur at

$n = 1$ and $n = 3$, and their fitted radial spectra agree with Eq. (S68) to first order. The pixelwise centre refinement returns $(\widehat{\delta}_z, \widehat{\delta}_y) = (6.7530 \times 10^{-5}, 2.62555 \times 10^{-3})$ and reduces the fitted odd-power fraction to 1.2879×10^{-7}. It also recovers the three generating even-mode coefficients to the displayed numerical precision. The remaining q_z displacement is an interpolation-dependent numerical residual and produces small odd cosine powers. Starting from this result, the additional modal refinement of Eq. (S37) returns the total shift $(5.2\times10^{-12}, 2.625000007\times10^{-3})$ and reduces the odd-power fraction to 5.95×10^{-21}. The example writes separate uncorrected, pixel-centred, and modal-refined standard overviews and a direct comparison of the induced $I_1^s(q)$ and $I_3^s(q)$ spectra with their first-order predictions.

The regression test `tests/test_stage_5_regression.py` fixes this noise-free construction. It requires recovery of both centre coordinates to within 0.02 nominal pixels, identifies $n = 1, 3$ as the two strongest raw odd sine orders, and requires the centre correction to suppress the odd-power fraction by more than three orders of magnitude while recovering the known even coefficients. It additionally requires the bounded modal refinement to recover the imposed shift to 10^{-8} in both reciprocal-space coordinates and to reduce the final odd-power fraction below 10^{-16}.

c. Interpretational boundary. This test demonstrates that a small origin error is sufficient to create the kind of coherent odd-mode structure seen in an uncentered detector map. If the structure disappears after a small, independently plausible centre correction and the recovered $n = 1, 3$ spectra follow the translation formulas, a geometrical origin is strongly supported. The inference remains conditional on the physical parity assumption used by the centering objective. A genuinely parity-odd contribution can also be reduced by that operation and must not be declared instrumental merely because it disappears. Minimizing all odd modes is therefore not a more model-independent operation than pixelwise parity centering; it is a useful numerical cross-check when the parity constraint has already been established physically. Stability against reasonable changes of $\mathcal{K}_{\mathrm{cen}}$, annulus, and shift bound should be checked for measured data. Moreover, the present fit treats the recovered centre as fixed; its estimation uncertainty and the interpolation correlations are not propagated into the reported modal covariance. Stage 5 is therefore a systematic-error verification and sensitivity test, not a calibrated hypothesis test that excludes all physical odd contributions.